

Master table

ci-tec | README | 24.09.2026

Use **Bio_O_Ton_Master.csv**, one row per recording. No separate ci-tec table is required. This is the latest recorded inventory, not a guarantee that processing is finished. Original metadata source: dawn-chorus-soundscape.csv.

Column	Meaning
dawn_chorus_id	Recording ID.
lon, lat	WGS84 coordinates in decimal degrees.
grid_100m_id, grid_10m_id	Assigned grid cells. A grid ID does not imply that LRT formation data are available.
datetime_local	German local date and clock time, with summer/winter time accounted for. No timezone suffix. Example: 2025-03-01 16:03:00.
inside_lrt_polygon	Recording point lies within an included LRT polygon.
grid_100m_has_majority_formation, grid_10m_has_majority_formation	Majority formation is available for this cell; different from a direct point/LRT intersection.

Grid embeddings, the LRT GeoPackage and grid reference data are separate products. Recording/bioacoustic flags do not certify their completeness.

Availability and issues

_exists: True means the corresponding files were recorded as present on LSDF. These flags describe the recorded inventory, not a live file check.

_has_issues: True means a problem or missing required data/check results. It can be True whether _exists is True or False.

Empty means no check result is available.

_issue_codes gives the reason. Multiple codes are separated by a pipe; details may follow a colon. **_has_issues is not a severity level or a universal exclusion rule.** Rules below concern the affected file/domain, not automatically the entire recording. Unknown codes or a True issue flag without codes require review.

Columns	Data
sound_exists, sound_has_issues	Audio file and audio checks.
sentinel_exists, sentinel_has_issues	Sentinel-2 files and technical checks.
weather_point_exists, weather_point_has_issues	Weather file for the recording and its checks.
weather_raster_hostrada_100m_exists, weather_raster_hostrada_100m_has_issues	HOSTRADA raster collection and its checks. These flags describe the whole collection and repeat for every recording.

Sound examples:

- True / False: file found, no issues reported.
- True / True: file found, but for example its duration is outside 59-61 seconds.
- False / True: file missing.

Sound codes and suggested use

sound_issue_codes	Meaning / consequence
duration_not_60s	Outside 59-61 seconds. Information ; decide in application-specific preprocessing. Not a general exclusion.
missing_file	File absent: unavailable.
empty_file, no_audio_stream_found, no_audio_frames_decoded	No usable audio in the checked file: exclude that file pending repair.
pyav_decode_failed, invalid_or_missing_sample_rate, invalid_or_missing_channel_count, invalid_duration, duration_unavailable	Technical check failed or a property could not be established: hold for review. Does not prove the original source is permanently corrupt.
multiple_audio_streams	Select the intended stream before use.
filename_does_not_match_<id>_audio_pattern	File-to-ID association needs clarification.
source_stat_failed, copy_failed, WinError	Access/copy problem; investigate connection and download before concluding source corruption.

Proposed audio filter: sound_exists=True and either sound_has_issues=False with no codes, or duration_not_60s as the only code. Do not overlook additional codes. This is a proposed usage rule for agreement, not an existing release approval.

Point weather codes and suggested use

Expected window: ten preceding complete German calendar days plus the recording day, hourly. Normally 264 rows; across spring DST 263, across autumn DST 265. The repeated local autumn hour is valid; a timezone-free timestamp alone cannot distinguish its two UTC instants.

weather_point_issue_codes	Meaning / consequence
missing_file	No weather file. Cause is not established by this code.
missing_value	At least one required value is missing. Potentially usable in part: check variables/hours needed by the application. Do not fill values blindly.
recording_date_changed_recheck_required	Recording date changed while previous results were retained. Temporal association is unconfirmed: hold until rechecked.
unexpected_time_interval	Full timestamp sequence and multiplicities differ from the expected window. May mean missing, extra or shifted hours. Correct DST transitions are accepted by the current inventory. Check/correct before use.
unexpected_time_window, unexpected_row_count	Wrong boundaries or row count: check/correct before use.
duplicate_datetime	More repeats than allowed by the expected window, not the legitimate repeated autumn hour. Check/correct before use.
missing_required_column, missing_datetime_column, unparseable_datetime, invalid_metadata_datetime	Structure or time basis is unreliable: hold until clarified.
empty_file, read_error	Empty or unreadable; an exception suffix may follow read_error. File currently unusable.
implausible_value	Temperature/humidity range check failed. Investigate values and source before deciding usability.

Strict weather filter: weather_point_exists=True and weather_point_has_issues=False with no codes. **Optional partial-data filter:** permit missing_value as the only code, then check required values yourself. Do not permit additional time/structure errors. New recordings do not necessarily lack files, and old recordings do not necessarily have NaN files.

Grid weather and Sentinel

Code / field	Meaning / consequence
weather_raster_hostrada_100m_*	Collection-wide status repeated across recordings. Many flagged rows do not mean that many independently corrupt raster files.
missing_raster, all_nodata	Raster missing or containing only NoData: affected product unavailable.
unexpected_shape, raster_structure_warning, raster_qc_read_error, qc_not_run	Raster structure/check needs review in the detailed raster QC.
Sentinel: missing_file, score_without_tif	Image missing; a score does not replace it.
Sentinel: quality_score_missing	Image may exist, but its quality assessment is missing.
Sentinel: empty_file, raster_read_failed	Image empty or unreadable: exclude pending repair.
inventory_not_run, qc_not_run	Check results unavailable; not equivalent to passing QC.

Weather and Sentinel do not automatically exclude a recording from audio/LRT-only work. Photos are not a filter criterion for the stated ci-tec priorities.

Processing completion is separate from quality

is_final does not currently exist. ready_for_general_analysis is not a ci-tec release flag: it requires unflagged sound, weather and Sentinel together and also rejects pure duration warnings.

For agreement: completion should mean that processing and checks for the required products and a specified input/code version are finished, with no outstanding repair. Missing data and application suitability remain separate questions. An irreversible True is only defensible for a frozen, versioned release. A new data version must be able to expose newly discovered errors. No rows are removed from the shared master under the proposed rules.

Abstimmung mit ci-tec: bitte bestätigen

Vorschläge, noch keine bestätigte Freigabe. Gemeinsame Datei: **Bio_O_Ton_Master.csv**. Ein Ausschluss betrifft zunächst den jeweiligen Datenbereich, nicht automatisch die gesamte Aufnahme.

Entscheidung	Vorschlag zur Bestätigung
1. Audio auswählen	sound_exists=True; keine Issues oder ausschließlich duration_not_60s. Dauerabweichung ist ein Hinweis. Leere, nicht dekodierbare oder technisch ungeklärte Audiodateien bis zur Klärung ausschließen.
2. Wetter mit Lücken	missing_value als einzigen Code anwendungsspezifisch zulassen: benötigte Variablen und Stunden prüfen. Alternativ nur vollständig ungeflaggtes Wetter nutzen. Keine automatische Auffüllung.
3. Zeit- und Strukturfehler	unexpected_time_interval, unexpected_time_window, unexpected_row_count, duplicate_datetime, recording_date_changed_recheck_required und Lese-/Strukturfehler bis zur Korrektur ausschließen. Korrekte Sommer-/Winterzeitwechsel sind kein Fehler. Fehlende Dateien sind nicht verfügbar.
4. Benötigte Produkte und Abschluss	Für reine Audio-/LRT-Auswertung fehlendes Wetter oder Sentinel nicht pauschal ausschließen; Fotos sind kein Auswahlkriterium. LRT-Zuordnung separat prüfen. Abschluss pro benötigtem Produkt und festgelegter Version definieren; kein pauschales is_final.

Zusammengeführter Stand vom 24.09.2026: 111.866 Aufnahmen; 6.430 mit 100-m-Mehrheitsformation, 1.154 direkt innerhalb eines LRT-Polygons und 2.211 mit 10-m-Mehrheitsformation. Aufnahmezeiten bleiben unverändert in deutscher Ortszeit. Beim Punktwetter tragen 2.615 Aufnahmen ausschließlich missing_value; die zuvor gemeldeten Wetter-Zeitfehler sind in diesem geprüften Stand bereinigt. Fehlende Werte bleiben fehlend.

Offene Festlegung für ci-tec: Welche Wettervariablen und Stunden müssen für die konkrete Auswertung vollständig sein? Die oben erläuterten Issue-Codes bleiben erhalten; es wird keine zweite gefilterte Master erzeugt.